

UrbanGazeVis

A visualization system for analyzing eye-tracking data
on urban safety perception

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Motivation & gap

Motivation & gap · UrbanGaze data · UrbanGazeVis · Results · Limitations

Does this street look safe or unsafe?



Does this street look safe or unsafe?



→ Safe?

Does this street look safe or unsafe?



Safe?



Unsafe?

Does this street look safe or unsafe?



→ Subjective

What is urban safety perception?



But what visual elements are relevant?



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How has urban safety perception been measured?

1

Crowdsourcing with Street View



2

Desktop eye-tracking



3

Immersive and mixed reality



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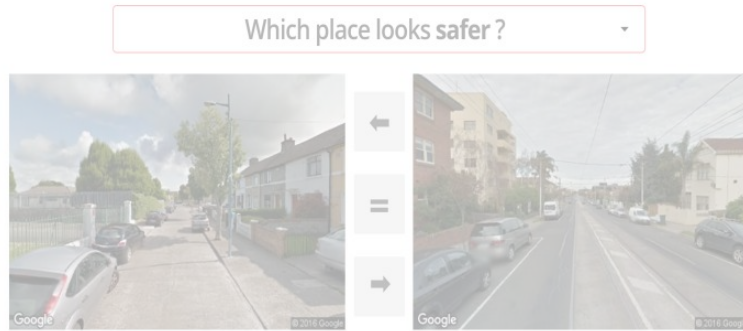
Immersive and mixed reality



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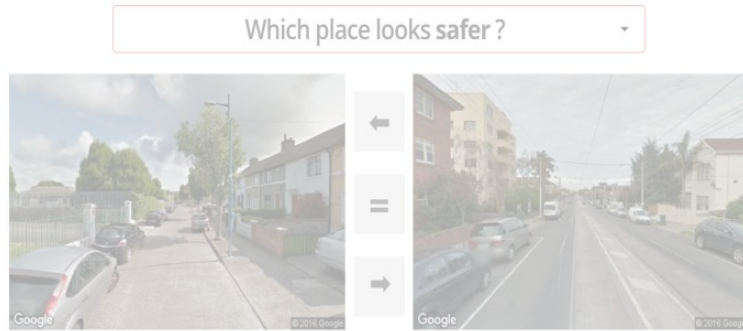
Immersive and mixed reality



How has urban safety perception been measured?

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What we still cannot answer



Which

parts of the street do people look at?



When

do they look at them: at the start, at the end, or again and again?



How

do people look differently at safe and unsafe streets?

What we still cannot answer



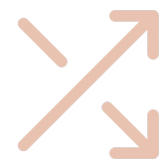
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Three gaps

- 1** No standard protocol for head-mounted gaze collection.
- 2** No dataset linking gaze, semantics, disorder cues and ratings.
- 3** No visual analytics tool for spatiotemporal, semantic gaze analysis.

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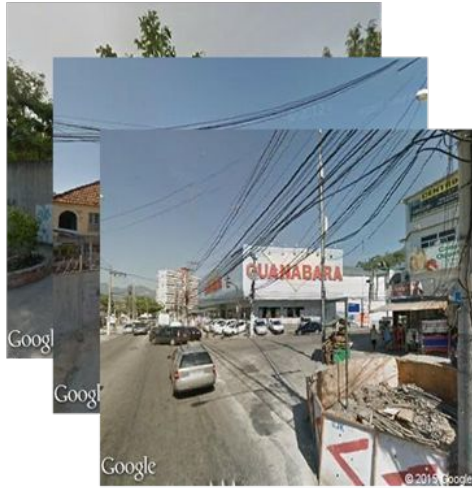
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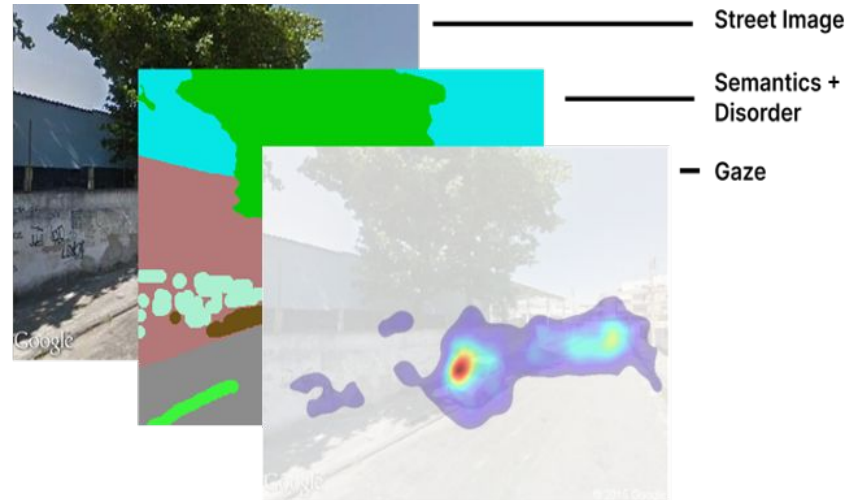
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Three contributions



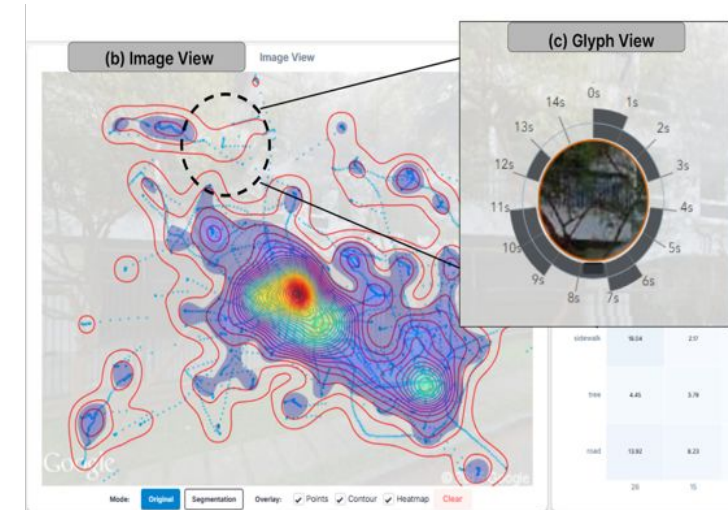
Protocol

Head-mounted gaze collection



Dataset

UrbanGaze



Tool

UrbanGazeVis

Three contributions



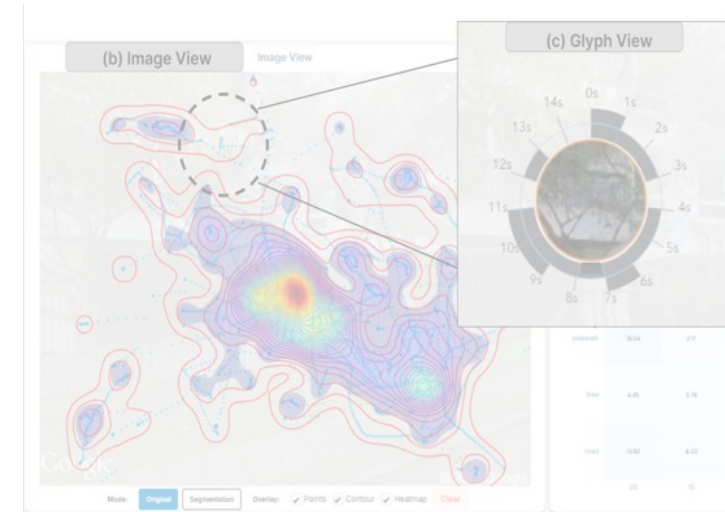
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Dataset

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Tool

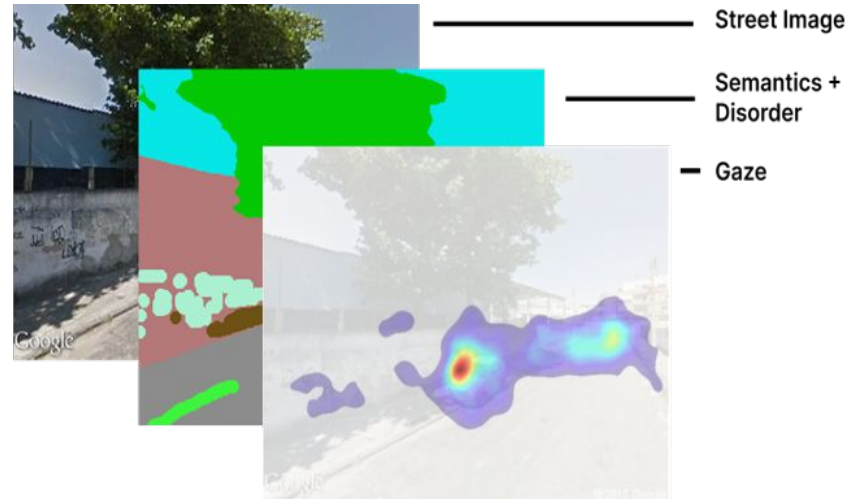
UrbanGazeVis

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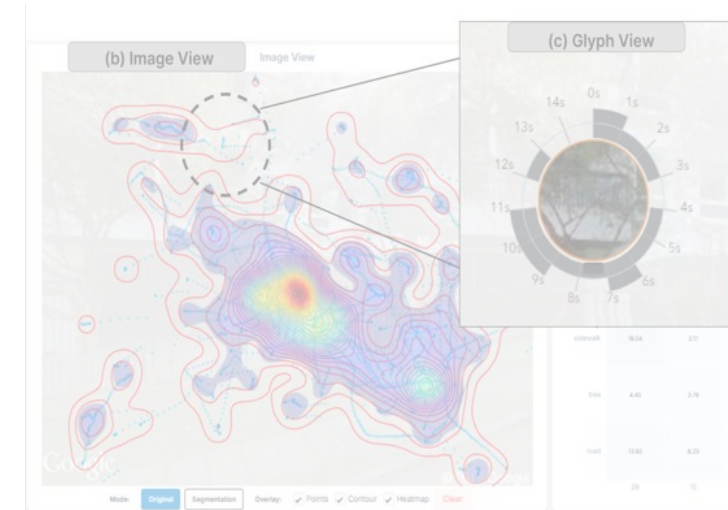
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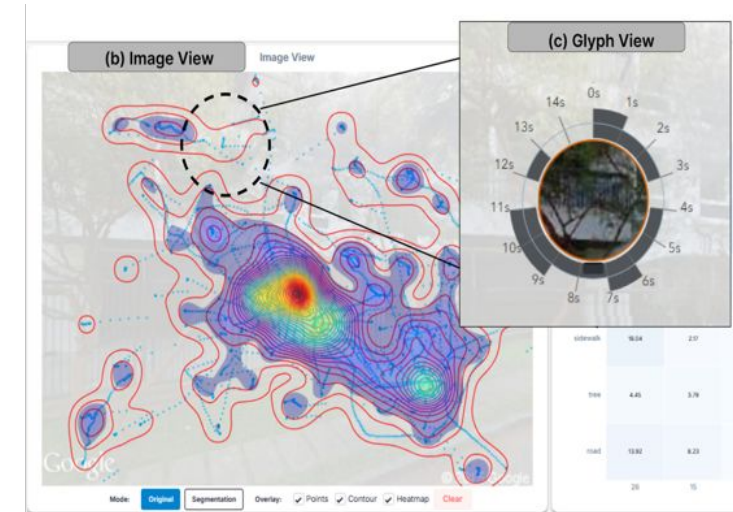
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UrbanGaze



Tool

UrbanGazeVis

UrbanGaze data

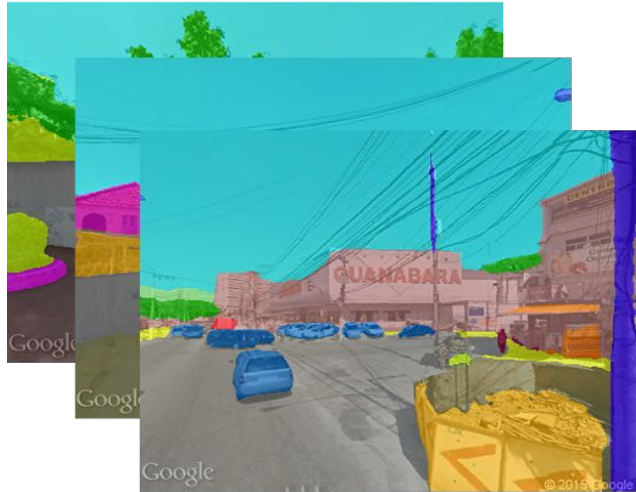
Contributions 1 and 2: collection methodology and dataset

Motivation & gap · **UrbanGaze data** · UrbanGazeVis · Results · Limitations

Dataset construction

1

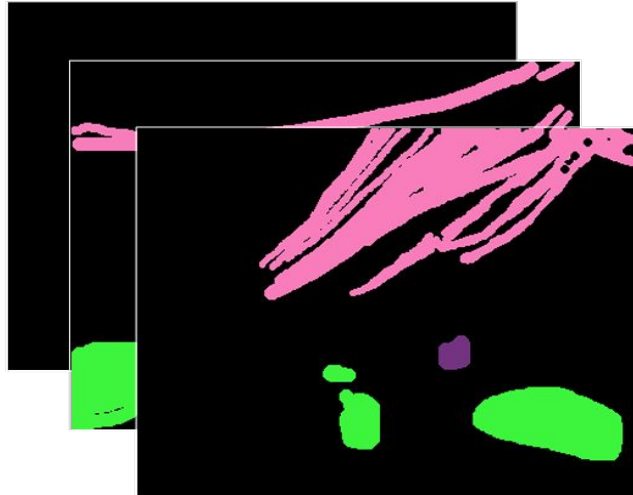
Stratified image selection



150 street views of Rio, balanced by stratified sampling

2

Semantics and disorder cues



A disorder layer: graffiti, cables, damaged walls, garbage

3

Image assignment

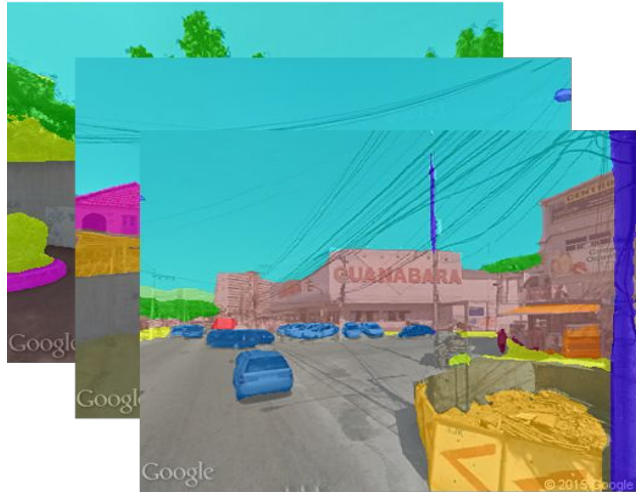


50 images per participant • 10 ratings per image

Dataset construction

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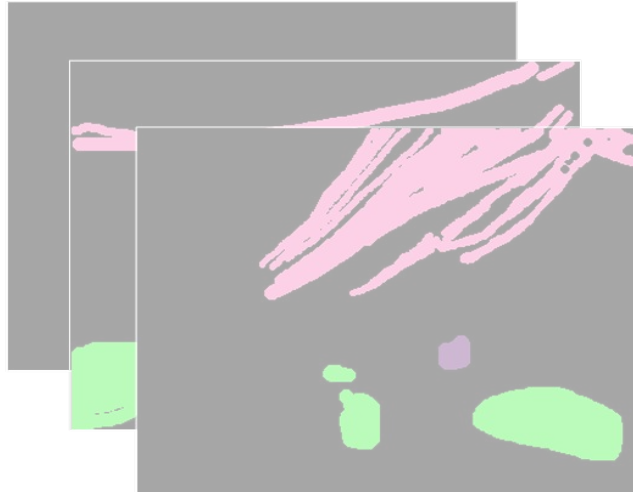
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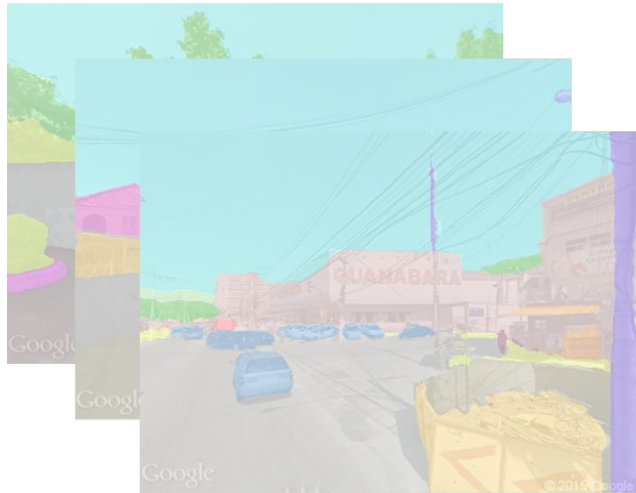


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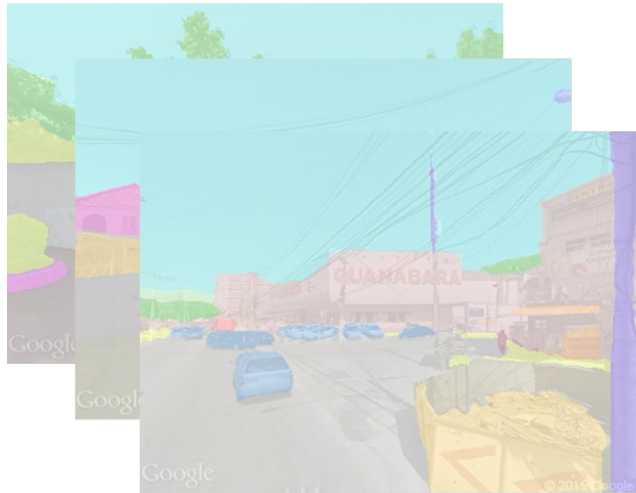


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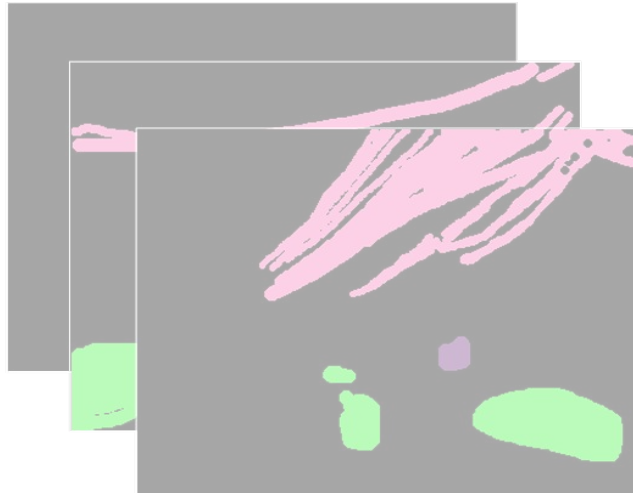
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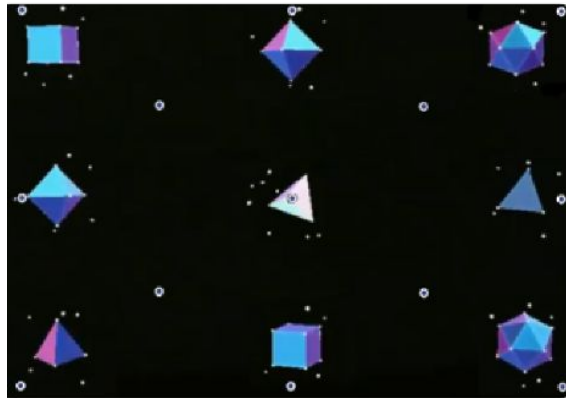


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Data Collection

1

Calibration



The HoloLens eye tracker is calibrated for each participant.

2

Familiarization



Participants see two examples — very safe and very unsafe — to understand the scale.

3

Main task

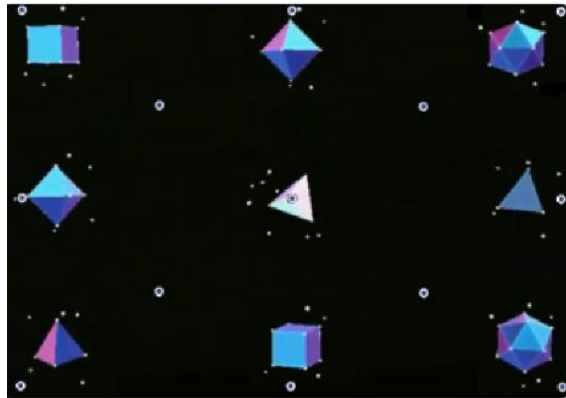


Each participant sees 50 images for 15 seconds, with a 5-second interval, and gives a score from 1 to 10.

Data Collection

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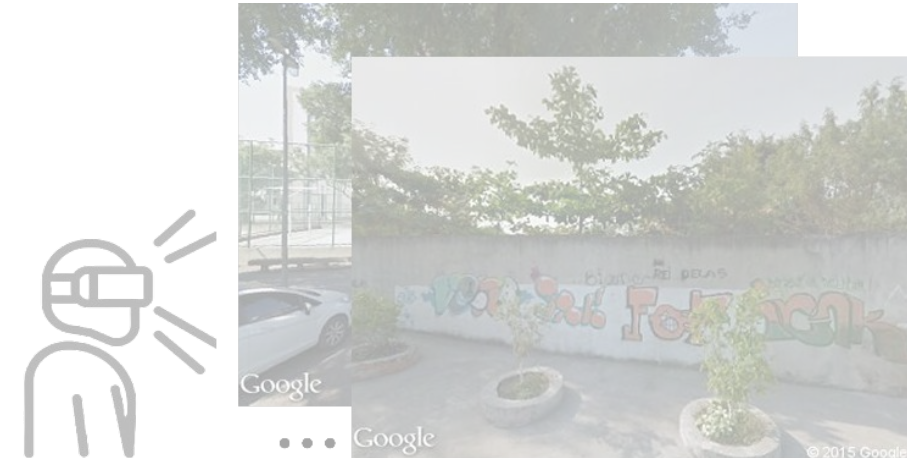
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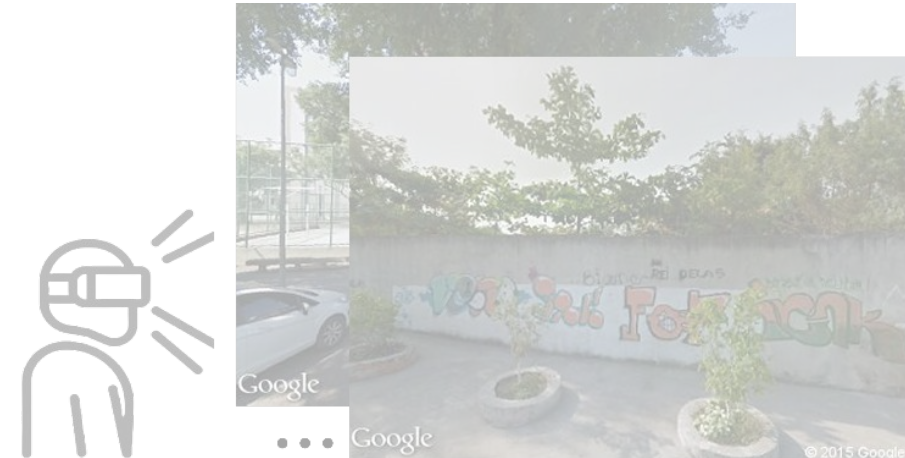
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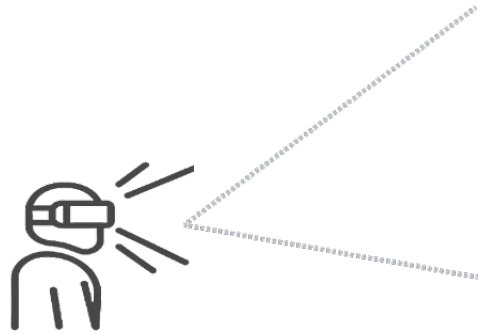
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The UrbanGaze dataset



30

participants



150

street-view images



50

per participant



**Safety
score**

10

ratings per image

Total attention is biased by object size



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Attention intensity corrects the area bias

$$\text{Attention intensity} = \frac{\text{Time looking at that class}}{\text{Fraction of the image that the class covers}}$$

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		Total attention	Attention intensity
Building	50% of the image	5 s	$5 / 0.50 = 10$
Car	1% of the image	5 s	$5 / 0.01 = 500$

UrbanGazeVis

Contribution 3: an interactive visual analytics system

Motivation & gap · UrbanGaze data · **UrbanGazeVis** · Results · Limitations

Design Goal

- 1 Support image- and participant- centric analysis**
- 2 Characterize spatiotemporal gaze behavior**
- 3 Relate gaze, semantics, and safety judgments**

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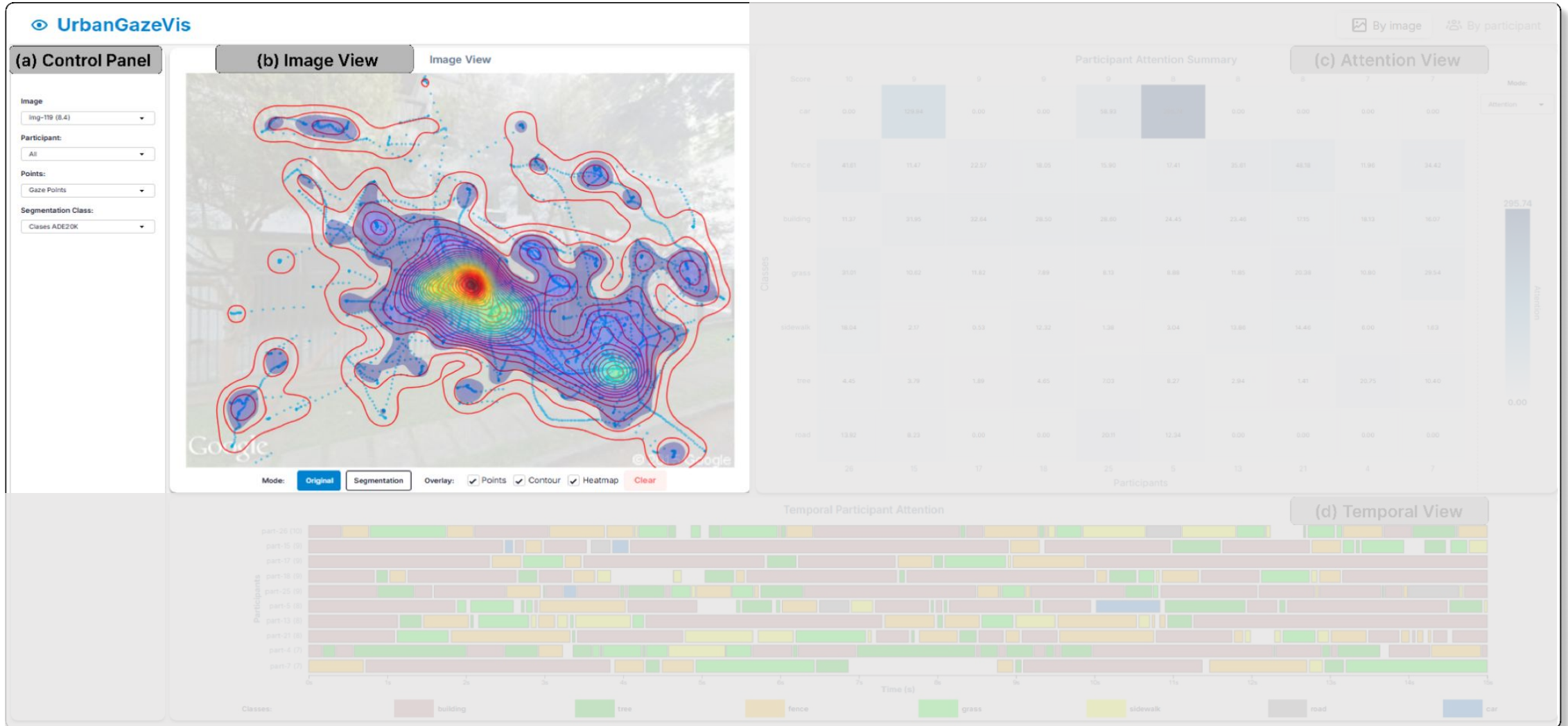
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UrbanGazeVis: By-Image component



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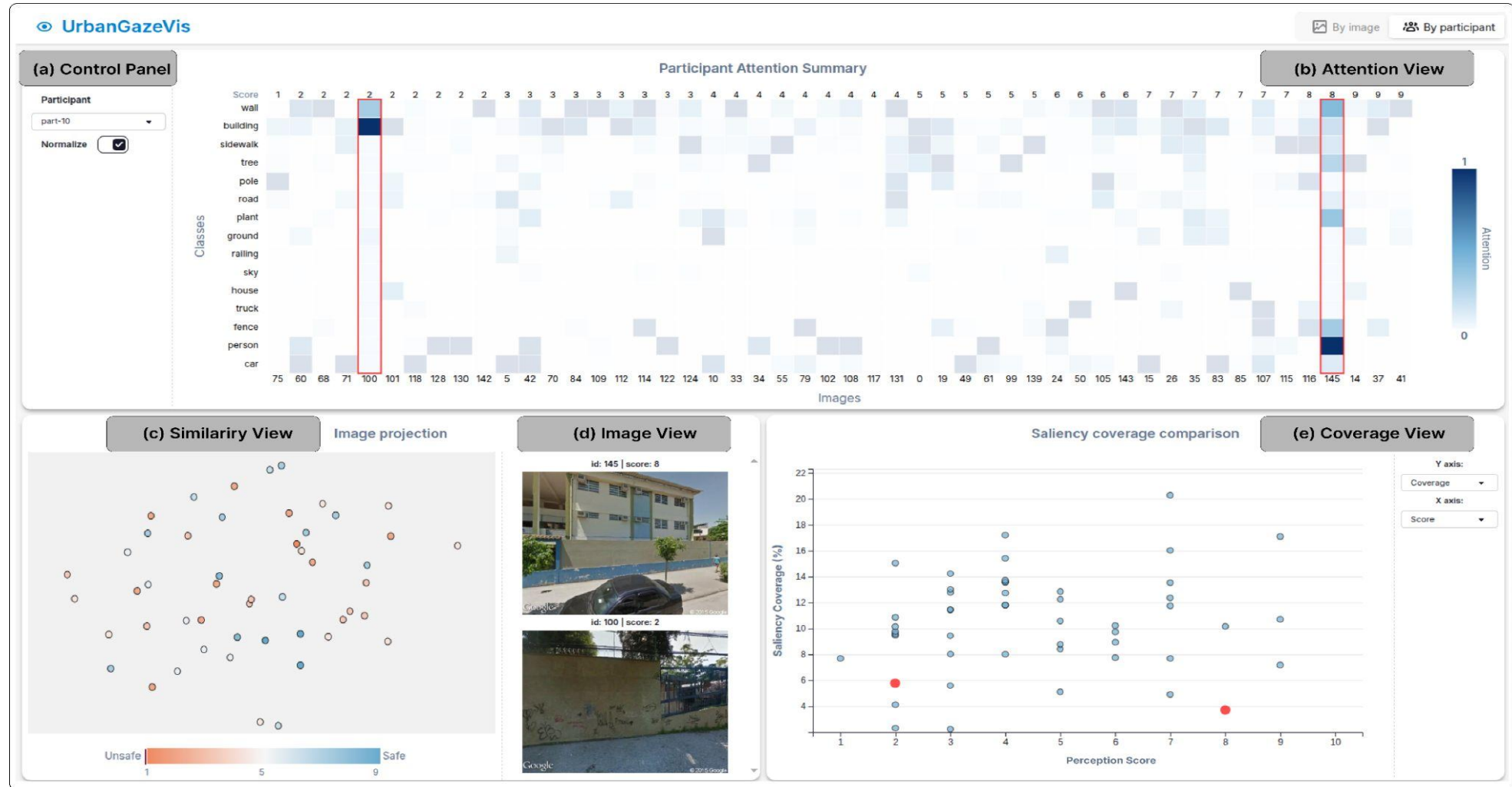
UrbanGazeVis: By-Image component



UrbanGazeVis: By-Image component



UrbanGazeVis: By-Participant component



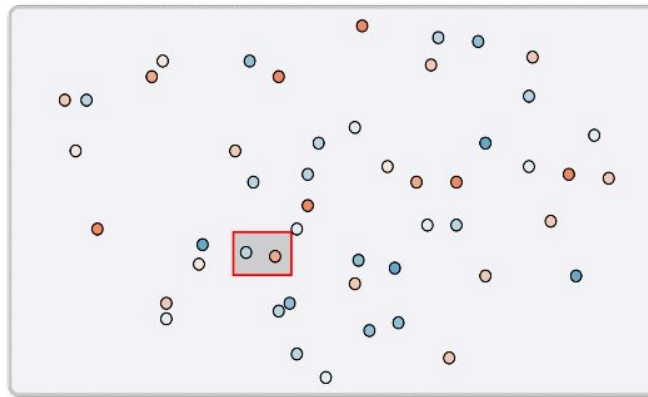
Results

Usage scenarios · statistical model · user study

Motivation & gap · UrbanGaze data · UrbanGazeVis · **Results** · Limitations

Scenario 1: Why do visually similar scenes get different scores?

Participant 2

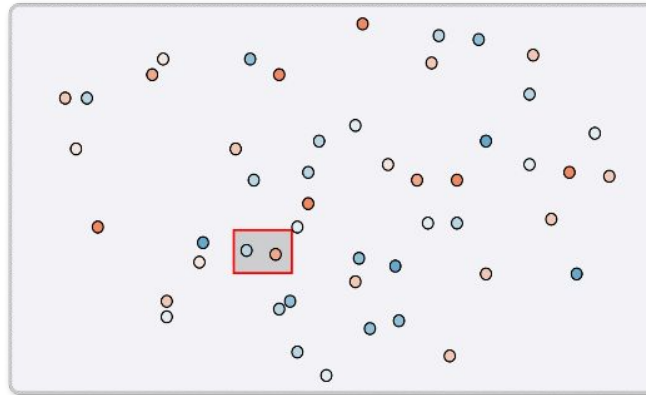


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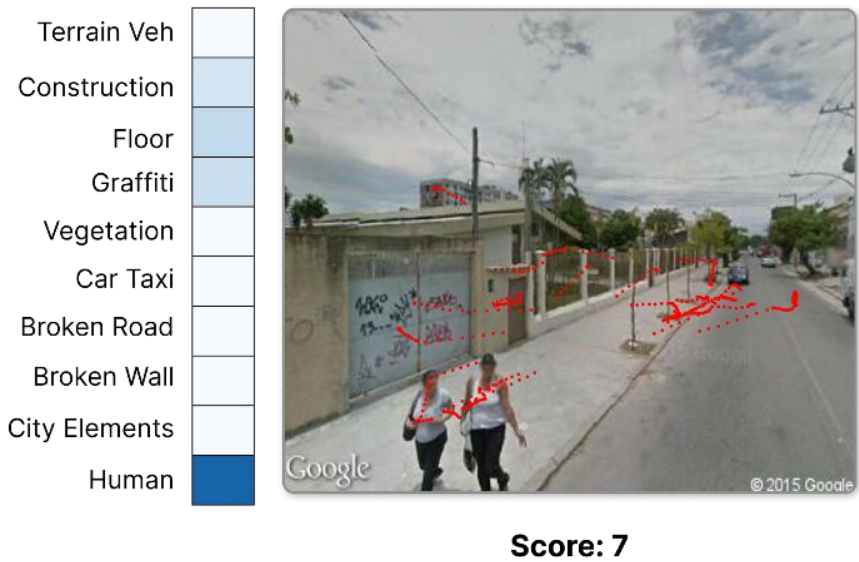
Score: 7

Participant 2

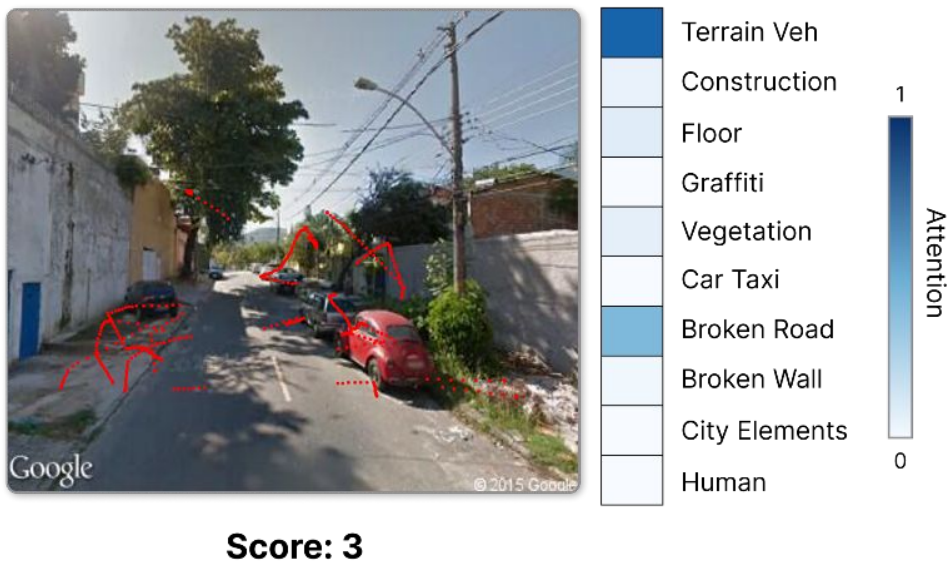
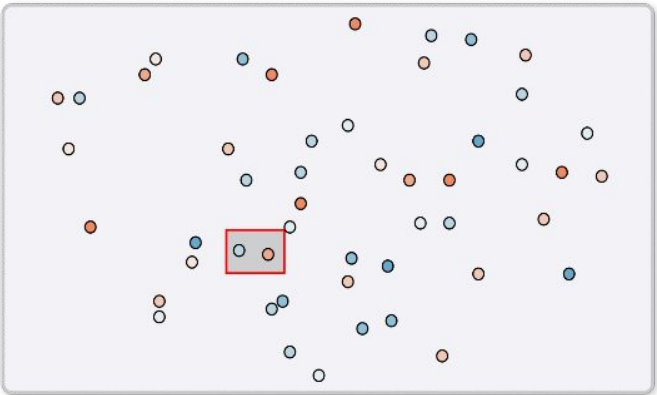


Score: 3

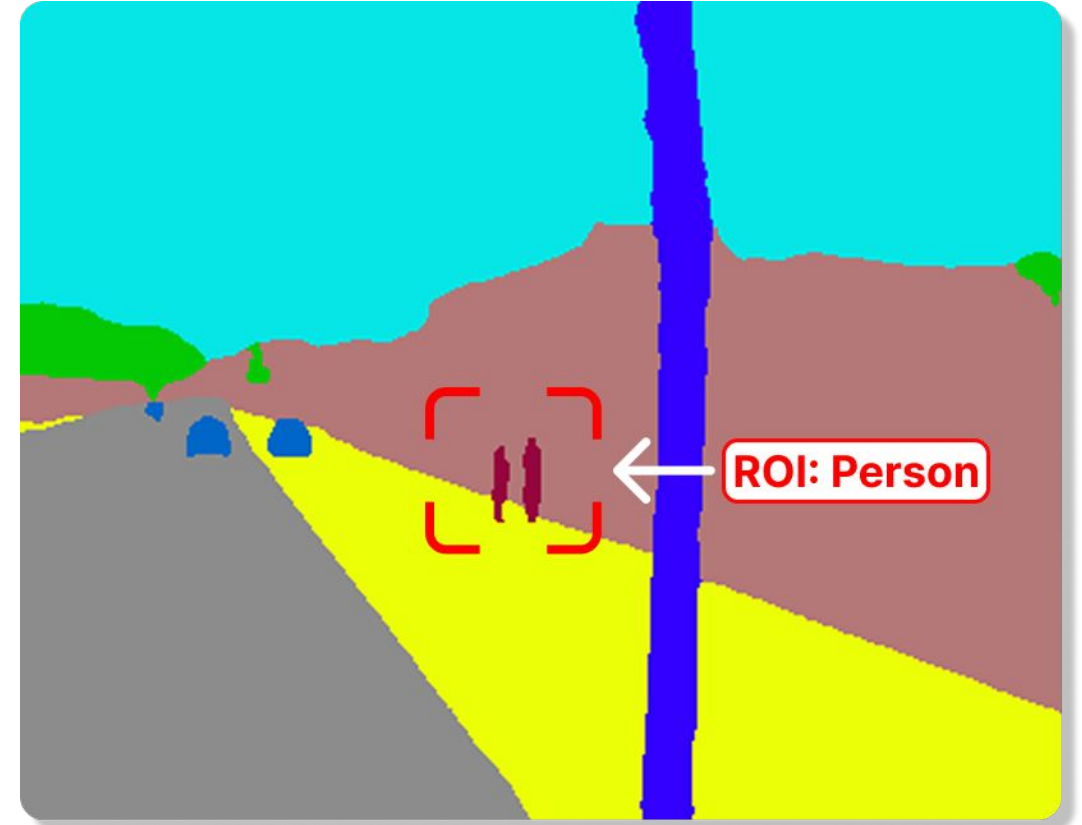
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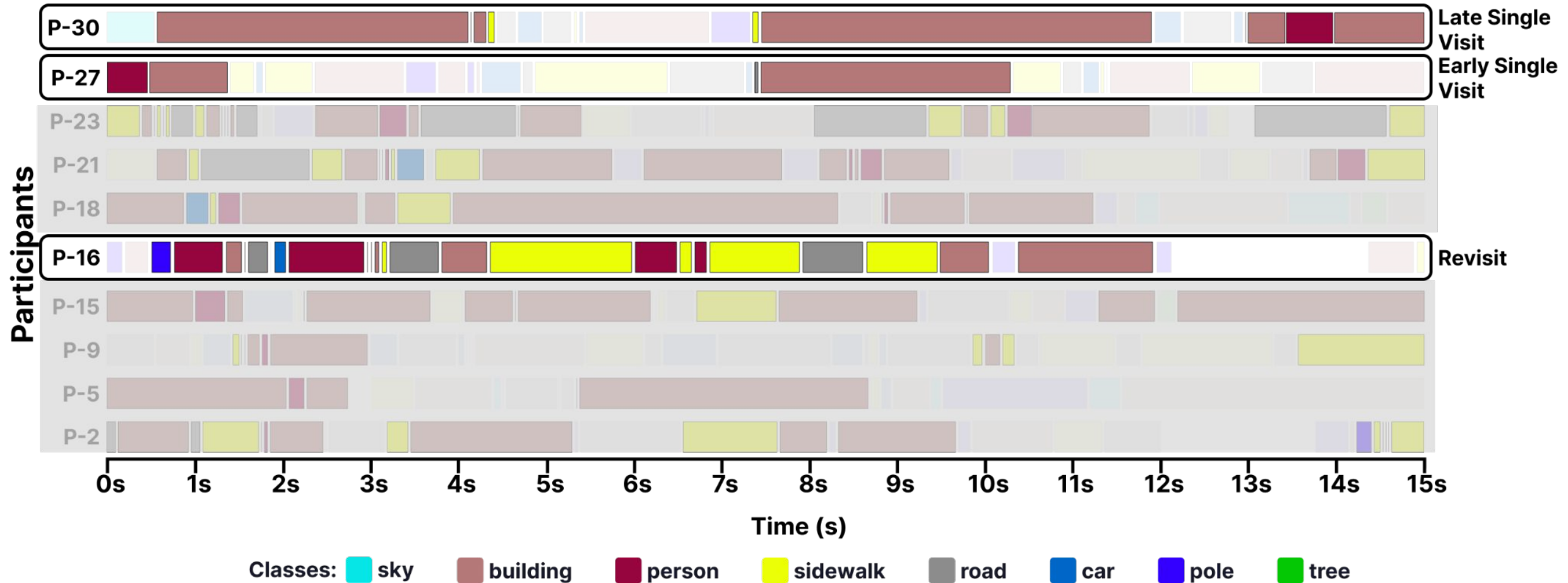
Participant 2



Scenario 2: Same hotspot, three ways of looking



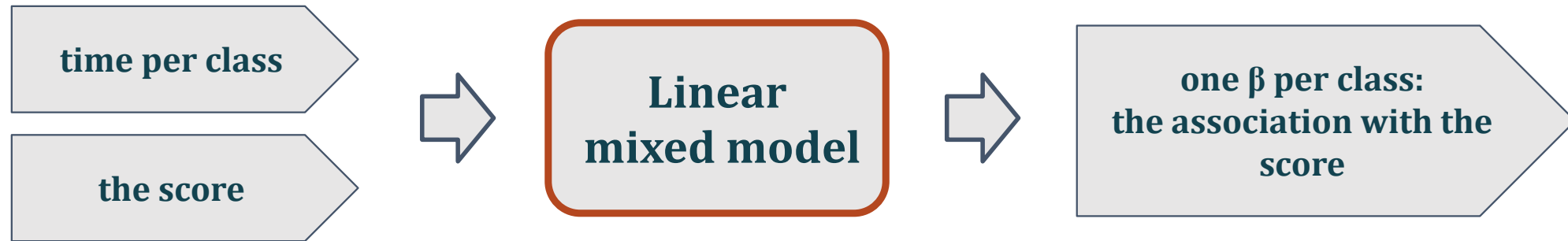
Scenario 2: Same hotspot, three ways of looking



Attention to disorder goes with lower safety



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LINKED TO LOWER SAFETY ($\beta < 0$)

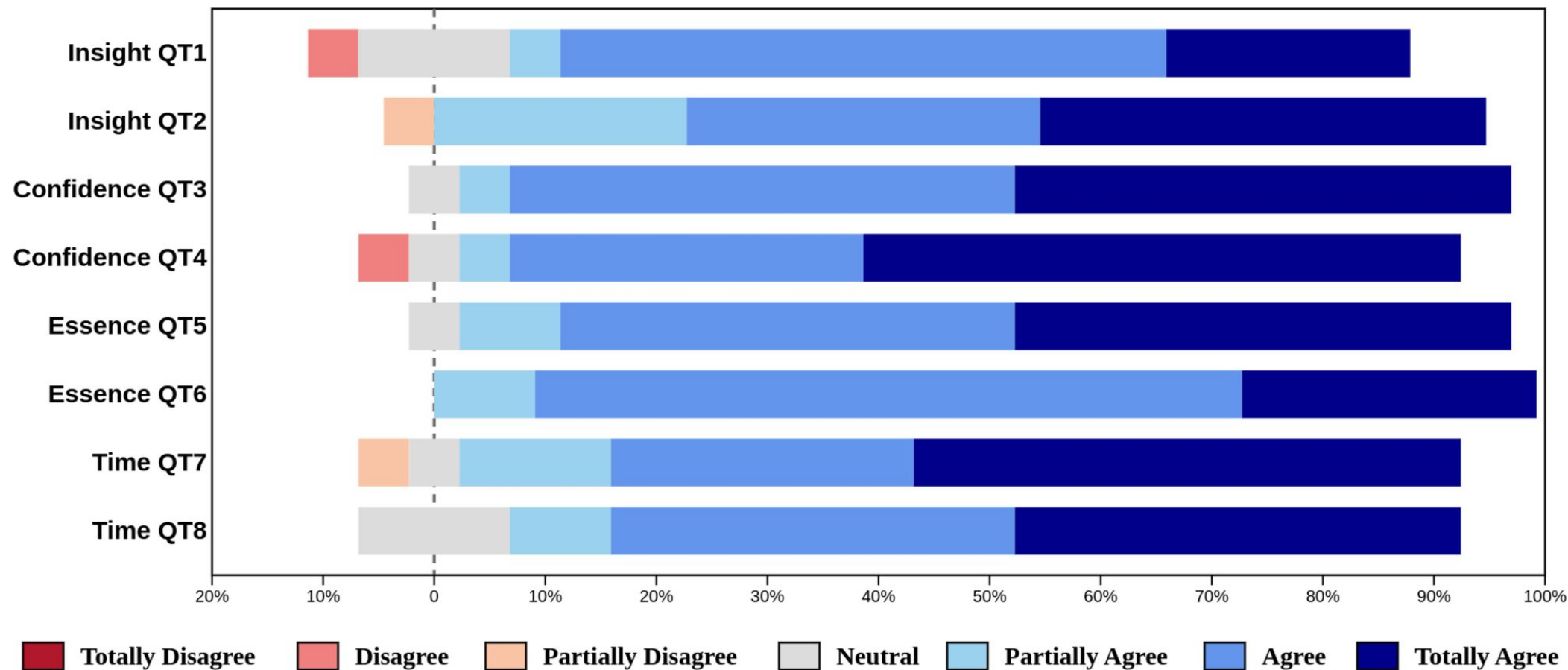
- Broken/Damaged Wall
- Damaged Pavement
- Graffiti
- Garbage Bag

LINKED TO HIGHER SAFETY ($\beta > 0$)

- Car / Taxi
- Vegetation
- Human

Evaluation with the ICE-T framework

22 participants, eight questions on a 7-point Likert scale. All four dimensions scored above the neutral midpoint.



Limitations

Motivation & gap · UrbanGaze data · UrbanGazeVis · Results · **Limitations**

Limitations

Population and setting

30 people, one institution, static images.

Scalability

The views are designed for 150 images and 30 participants.

Association, not causation

Attention to decay is associated with lower scores, but we cannot prove a cause.

Thank you

UrbanGazeVis

